Wainwright Community Permafrost Thaw Risk Assessment Project

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INTRODUCTION

The purpose of this project is to identify permafrost thaw risk areas for infrastructures and find out the places that are good for neighborhood extensions or relocations. 50-meter resolution grids were overlayed on the extends of Wainwright.



1. Infrastructure

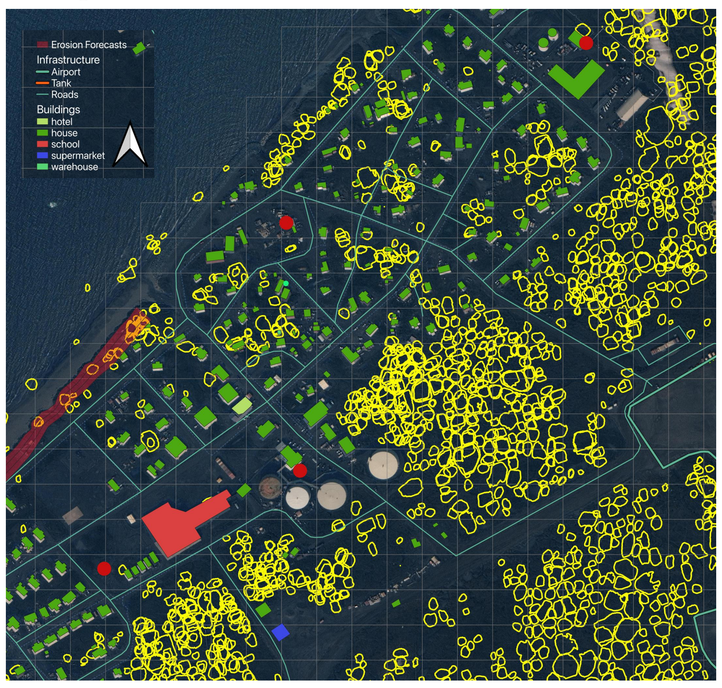
This dataset was built by **HABITAT** pipeline. HABITAT is a deep learning pipeline using **ResNet50-UNet++** architecture to map buildings and roads from high-resolution Maxar satellite imagery in Arctic communities [1].

2. Contaminated Sites

Contaminated sited data was downloaded from Division of Spill Prevention and Response - Contaminated Sites database [2].

3. Coastal Erosion Forecast

Erosion exposure assessment of infrastructure in Alaska coastal communities' data was published by Department of Geological Resources [3].



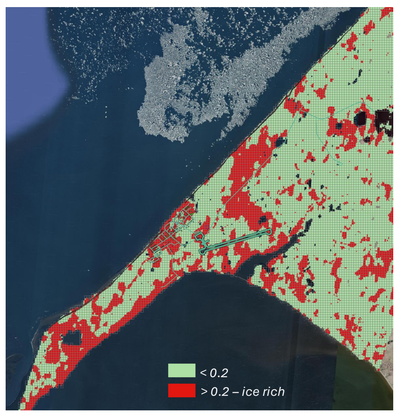
4. Ice Wedge Polygons (IWP)

This dataset maps high ice content permafrost regions using high resolution satellite imagery from 2001-2021. A CNN detects these polygons' boundaries and then produces shapefiles that are masked, re-projected circumference and width [4].

5. InSAR – Excess Ice

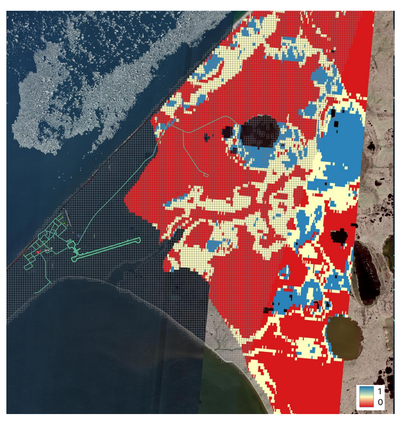
InSAR products created for PDG are vary. In this study, mean excess ice content was used. (Preliminary data produced by Simon Zwieback)

The values < 0.05 indicate ice-poor conditions, while values > 0.2 are ice-rich and thaw sensitive indicating potentially unstable.



6. Drained Lakes

Drained Lakes dataset provides lake area data for permanent and seasonal lakes in the Arctic from 1984-2021. This dataset was derived from Landsat imagery [5].



If the area has no drain lakes, it probably still ice. This is a risk. But, if the area has a drain lakes, it means there were permafrost, but it thawed.

7. Trough Ponds

Ponds form in the degrading troughs of ice wedge polygons. (Preliminary data, produced by Michael Pimenta).

METHODS

1 **Risk = Hazard x Exposure x Vulnerability**

Risk Index

1. **Hazard:** How likely is the ground here to destabilize? — Computed as: Weighted sum of physical features
2. **Exposure:** What is there that could be affected? — A cell with no infrastructure has zero exposure regardless of hazard
3. **Vulnerability:** If the ground fails here, how severe is the damage? — Quality of the infrastructure

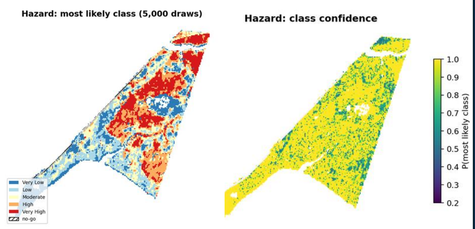
2 Create 50m x 50m Grids with Risk Index

The grids are like a chessboard if cells. Based on the x and y coordinates, rows and columns were figured out. Create a blank grid (array) for each feature. Fill in the values from the data.

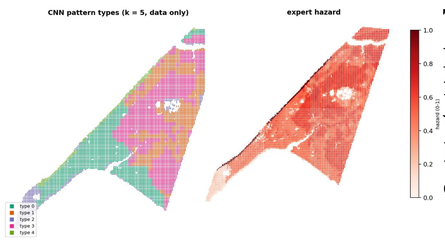
1	Drained lake basins *Bergstedt et al. 2021 · Landsat-8 + ArcticDEM*	HAZARD
2	Ice-wedge polygons *Witharana et al. 2023 · WorldView-2 CNN*	HAZARD
3	InSAR excess ice *Preliminary · S. Zwieback; >0.2 = ice-rich*	HAZARD
4	Trough ponds *Preliminary · NDWI · M. Pimenta*	HAZARD
5	Contaminated sites *Alaska Dept. of Environmental Conservation*	NO-GO
6	Coastal erosion forecast *Buzard et al. 2021*	NO-GO
7	Infrastructure footprints *HABITAT / 0.5 m Maxar · Manos 2024*	EXPOSURE

3 Stage 1: Weights and Robustness Check

To check the weight sensitivity, we run 5000 Monte Carlo draws. 99.5% of cells kept their hazard class.

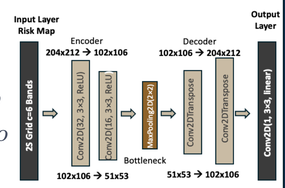


4 Stage 2: CNN for classification



The CNN does not produce a hazard map; it never sees hazard, weights or infrastructure. It finds patterns in the data.

Conv2D(32) from 6 Inputs: 3x3x6x32+32 = 1760
Conv2D(16) from 32: 3x3x32x16+16 = 4624
Conv2DTranspose(16) from 16: 3x3x16x16+16 = 2320
Conv2DTranspose(32) from 16: 3x3x16x32+32 = 4640
Conv2D(1) from 32: 3x3x32x1+1 = 289



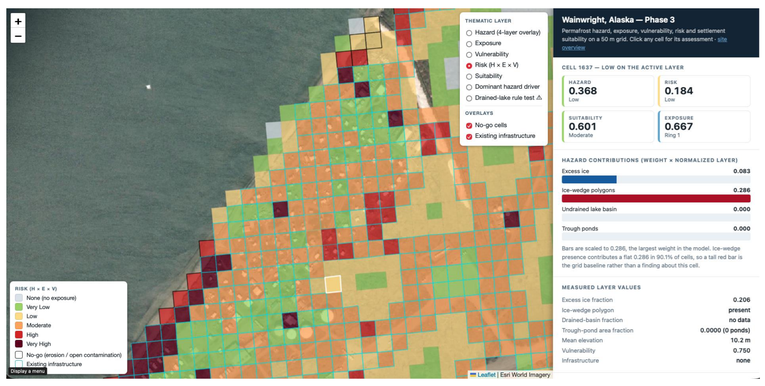
5 LLM Narratives (as a prototype)

Each cell prompt carries hazard, risk, no-go, the four layer contributions, the measured values, and the data-driven pattern type. The pattern type is never described as a predicted risk score. No generated narrative. Planners cannot act on text we have not produced. Evaluation with community planners is a Wainwright next step.

Sample cell · grid 5304 · test · highest risk

Generate a risk narrative for this 50 m grid cell in Wainwright, Alaska.
EXPERT MODEL OF RECORD (phase 3, hazard = weighted sum of four measured layers,
risk = hazard x exposurex vulnerability):
- **Hazard: 0.**777 (Very High, quintile 5 of 5)
- **Risk: 0.**777
- Exposure ring: ring 0 - the cell contains infrastructure (exposure 1.000, vulnerability 1.000)
- No-go override: none
HAZARD LAYER CONTRIBUTIONS (weight x measured value, they sum to the hazard....

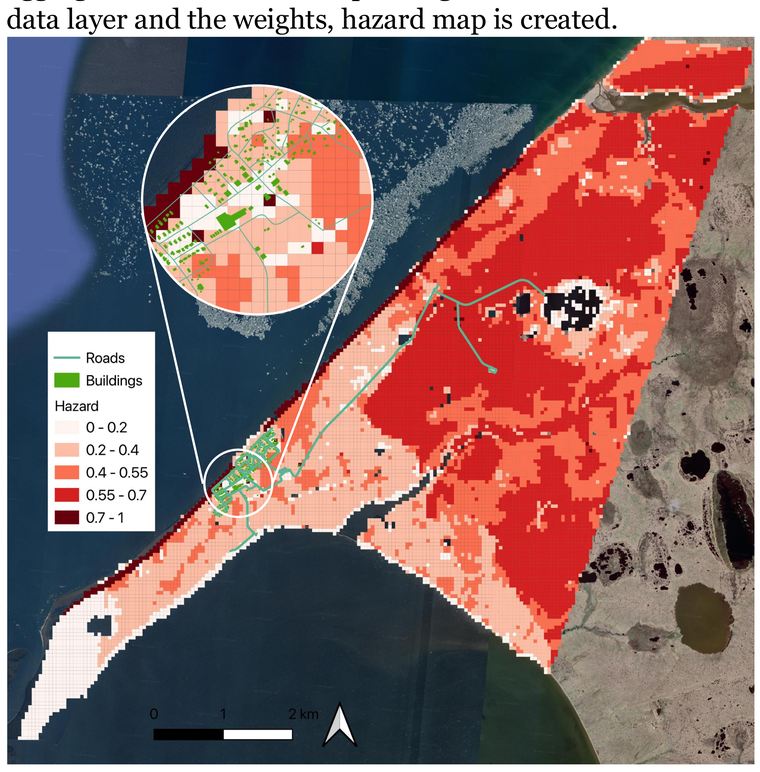
6 The Product



RESULTS

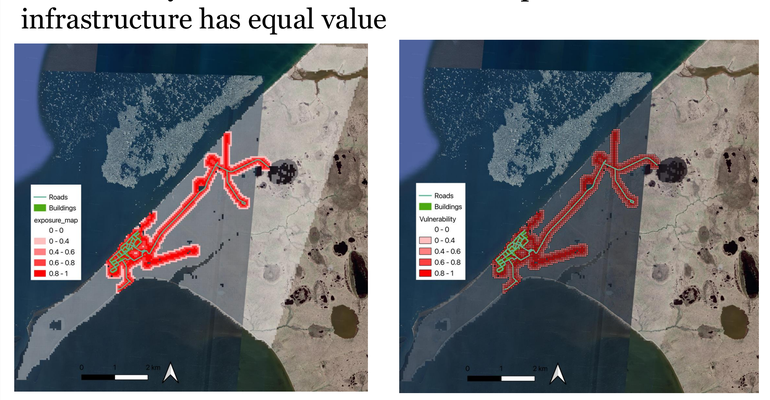
1. Hazard

Different weights were given to the hazard layers to aggregate hazard into 1. Depending on the existence of the data layer and the weights, hazard map is created.

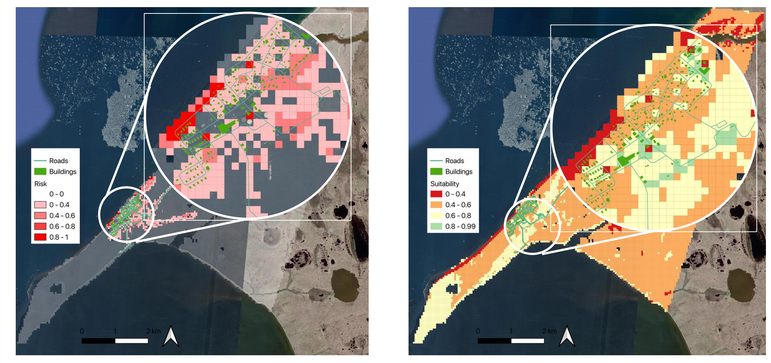


2. Exposure and Vulnerability

Vulnerability is defined similar to Exposure where each infrastructure has equal value



3. Risk and Suitability Map



CONCLUSION and FUTURE WORK

Permafrost thaw threatens 40% of the settlements in permafrost region by 2050 [7]. This pilot project develops a geospatial deep learning framework to assess permafrost thaw risk and identify potential relocation or neighborhood extension areas within a buffer zone of the Wainwright region.

Until the ground truth exists, every result here is robustness or convergence, not validation. Future work includes (1) An expert form for the physical layers, (2) Narratives that can be trusted, (3) Vulnerability that means something, (4) Ground-truth observations

ACKNOWLEDMENTS

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LEO NETWORK